

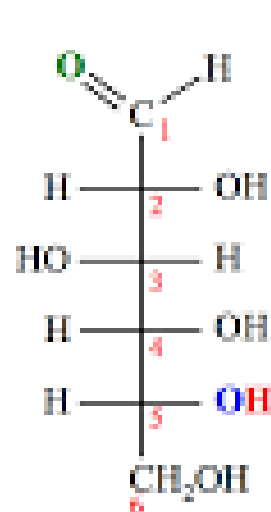
Biochemistry Lec.2

- Haworth Structure
- isomerism

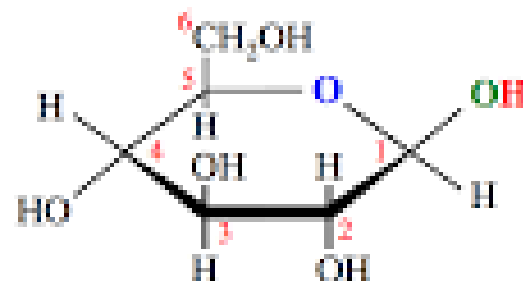
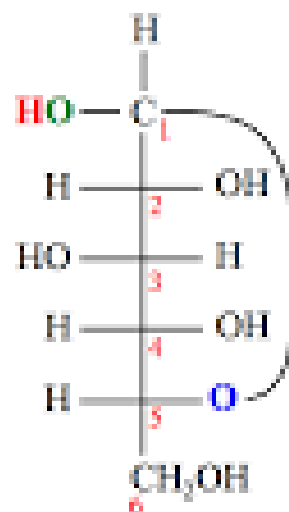


Haworth Structure

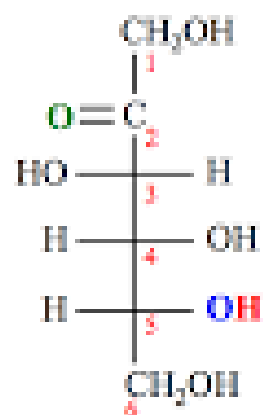
- ✓ **Common monosaccharides (pentoses and hexoses) occur in cyclic forms (closed chain)**
- ✓ **If the carbon chain is long enough (five or more carbons), the alcohol at one end of a monosaccharide can attack the carbonyl group (aldehyde or ketone) at the other end to form a cyclic compound (hemiacetal or hemiketal respectively).**
- ✓ **When a six-membered ring is formed, the product of this reaction is called a pyranose**
- ✓ **When a five-membered ring is formed, it is called a furanose**
- ✓ **The rings can open and re-close, allowing rotation to occur around the carbonyl carbon resulting in :
Two distinct configurations (α and β anomers) of the hemiacetals and hemiketals are present.**
- ✓ **The anomeric carbon is the carbonyl carbon (C1 of aldose or C2 of ketose).**



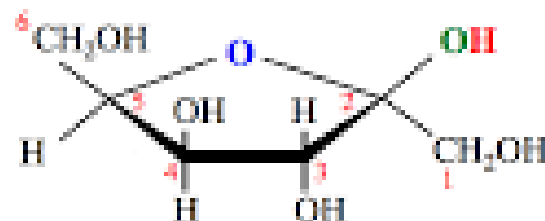
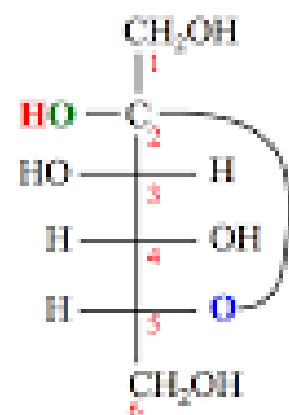
D-glucose
Fischer projection
(open-chain form)



D-glucose
Haworth projection
(cyclic form)



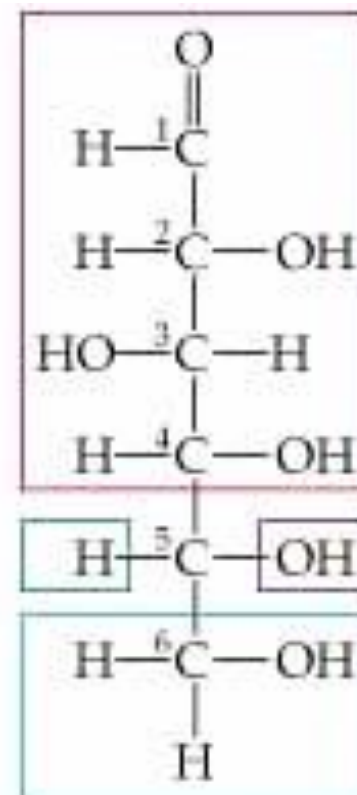
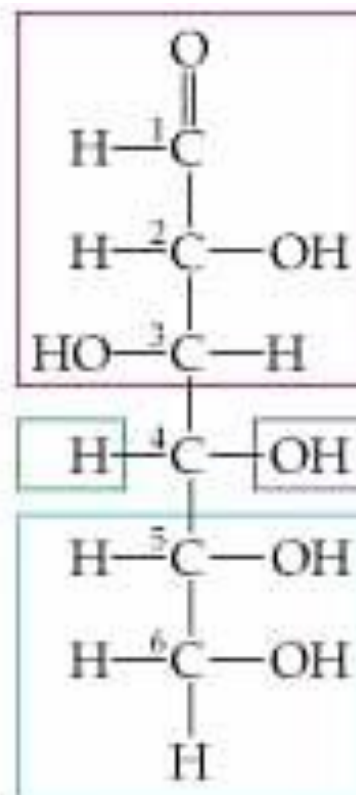
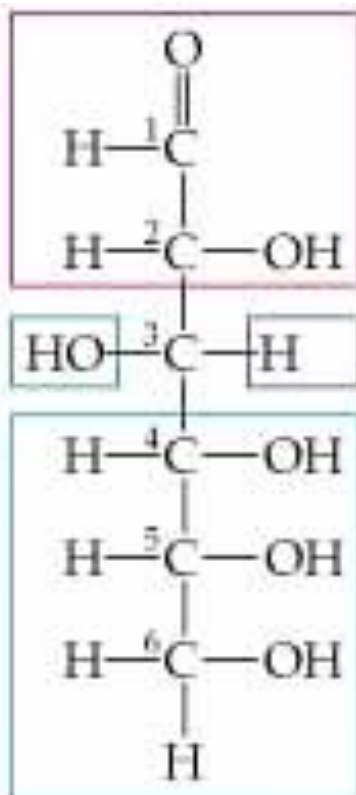
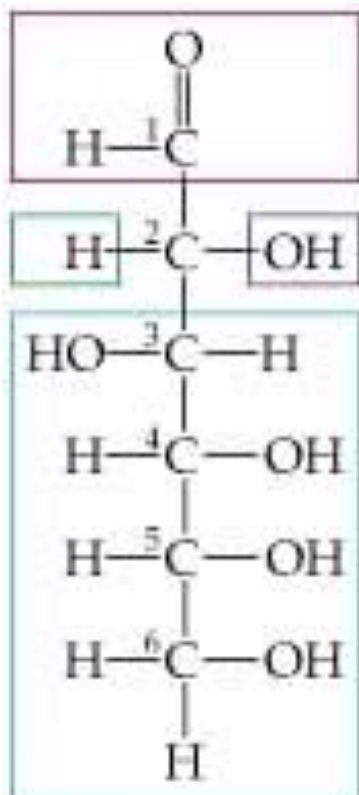
D-fructose
Fischer projection
(open-chain form)



D-fructose
Haworth projection
(cyclic form)

Asymmetric (chiral) carbon

- ✓ It is that carbon atom which is attached to 4 different groups or atoms.
- ✓ All carbohydrates except dihydroxy acetone have asymmetric carbon atom
- ✓ Any substance containing asymmetric carbon atom shows two properties; optical activity and optical isomerism



Optical activity

- ✓ The compounds having asymmetric carbon atoms can rotate the beam of plane polarized light and are said to be optically active.
- ✓ Sugar which can rotate the plane of polarized light to the right is called as dextrorotatory and is designated as **(d) or (+)**

Example: D- (d)-glucose or it is also known as **dextrose**.

- ✓ While the isomer which rotates the plane of polarized light to left is known as levorotatory, and is identified as **(l) or (-)**.

Example: D-(l)-fructose or it is also known as **levulose**.

- ✓ Optical activity of substance can be measured by polariscope

Racemic mixture: It is the mixture containing equal number of two optically active sugars, one is dextrorotatory and the other is levorotatory.

Resolution: It is the separation of optically inactive racemic mixture into its optically active substances.

Optical isomerism

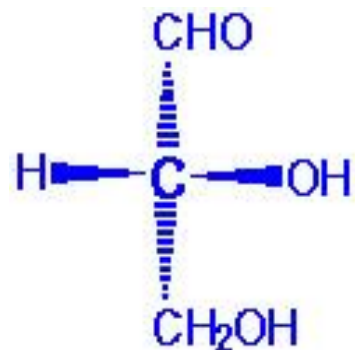
- Isomers are compounds having the same chemical formulae but differing in the arrangement of their atoms in space and having different physical properties are called isomers.
- The general formula for determining the number of stereoisomers is 2^n , where n is the number of chiral centers present in the molecule.

1- Enantiomers (stereoisomers) = D- and L- forms

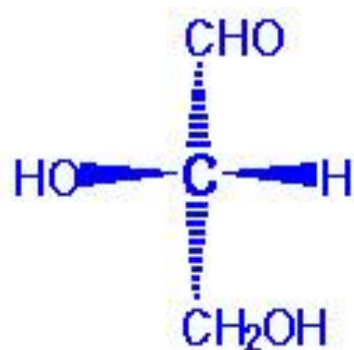
- Structures that are mirror images of each other and are designated as D- and L- sugars based on the position of -OH grp on the asymmetric carbon farthest from the carbonyl carbon.
- The stereoisomer in which the -OH extends to the right is called D-glyceraldehyde (from the Latin dexter= right), and the stereoisomer in which the -OH extends to the left is L-glyceraldehyde (from levus = left).

Enantiomers (stereoisomers)

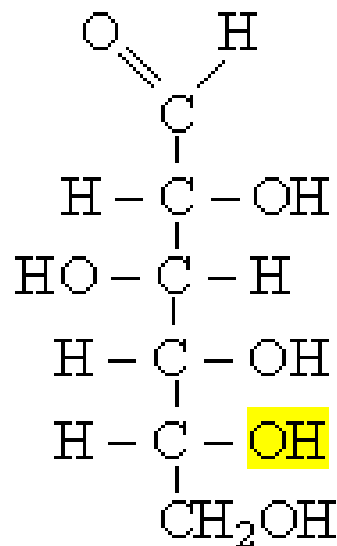
D - and L - isomers



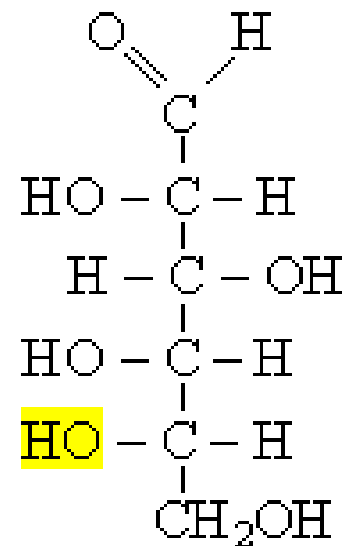
D-Glyceraldehyde



L-Glyceraldehyde

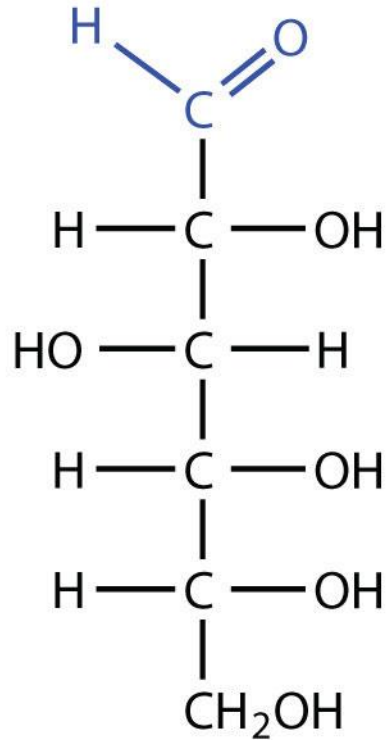


D-glucose

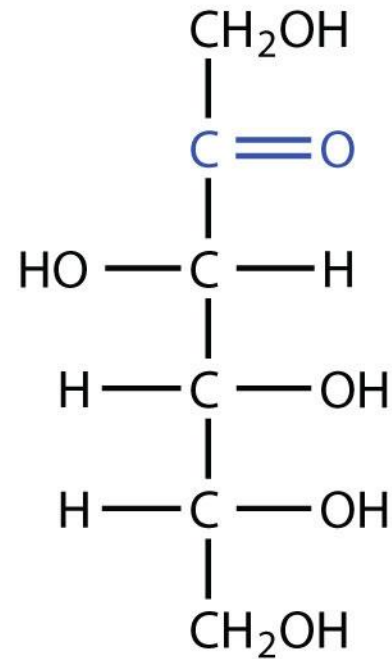


L-glucose

2- Aldo Keto isomers



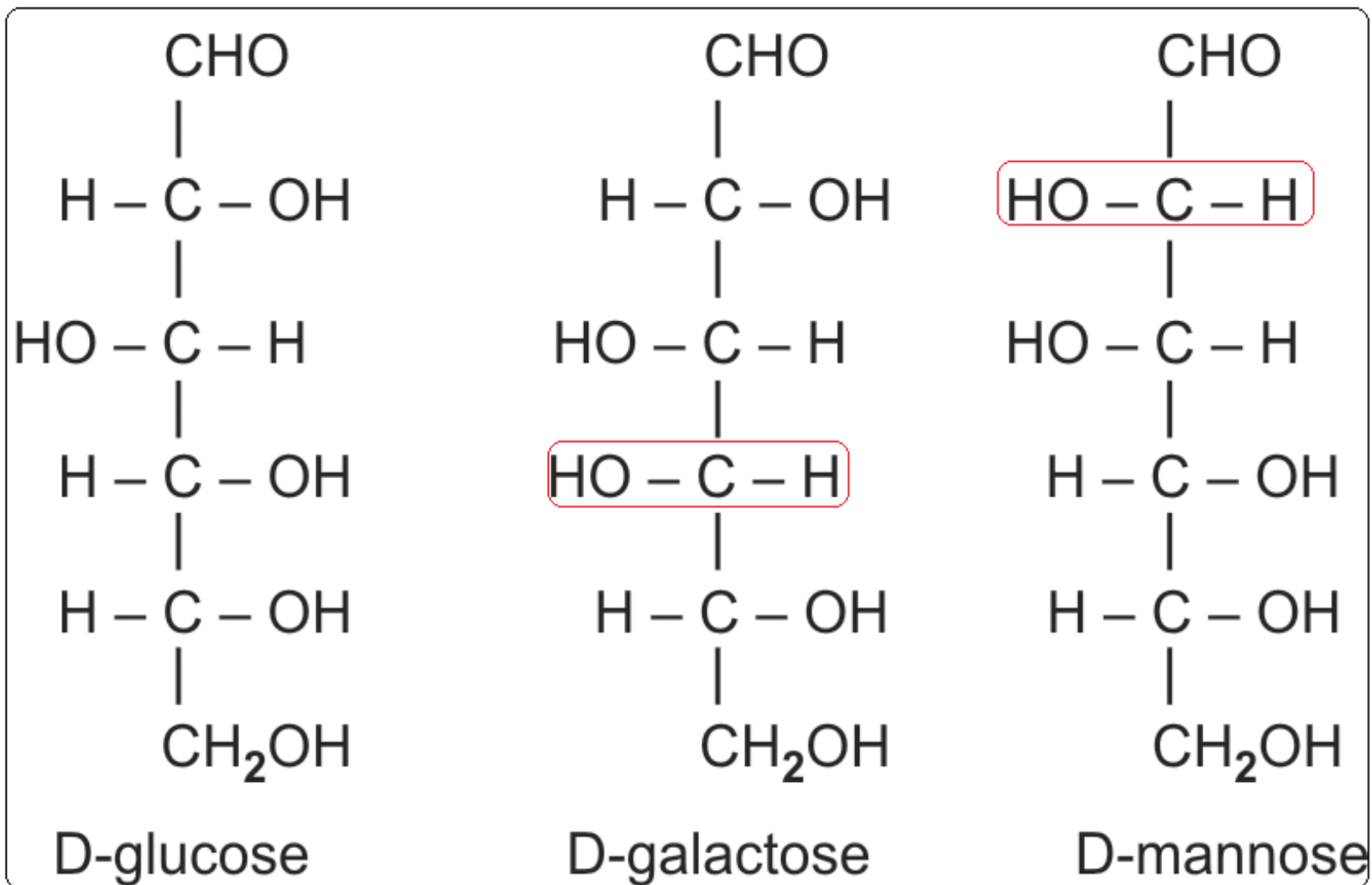
Glucose
(an aldohexose)



Fructose
(a ketohexose)

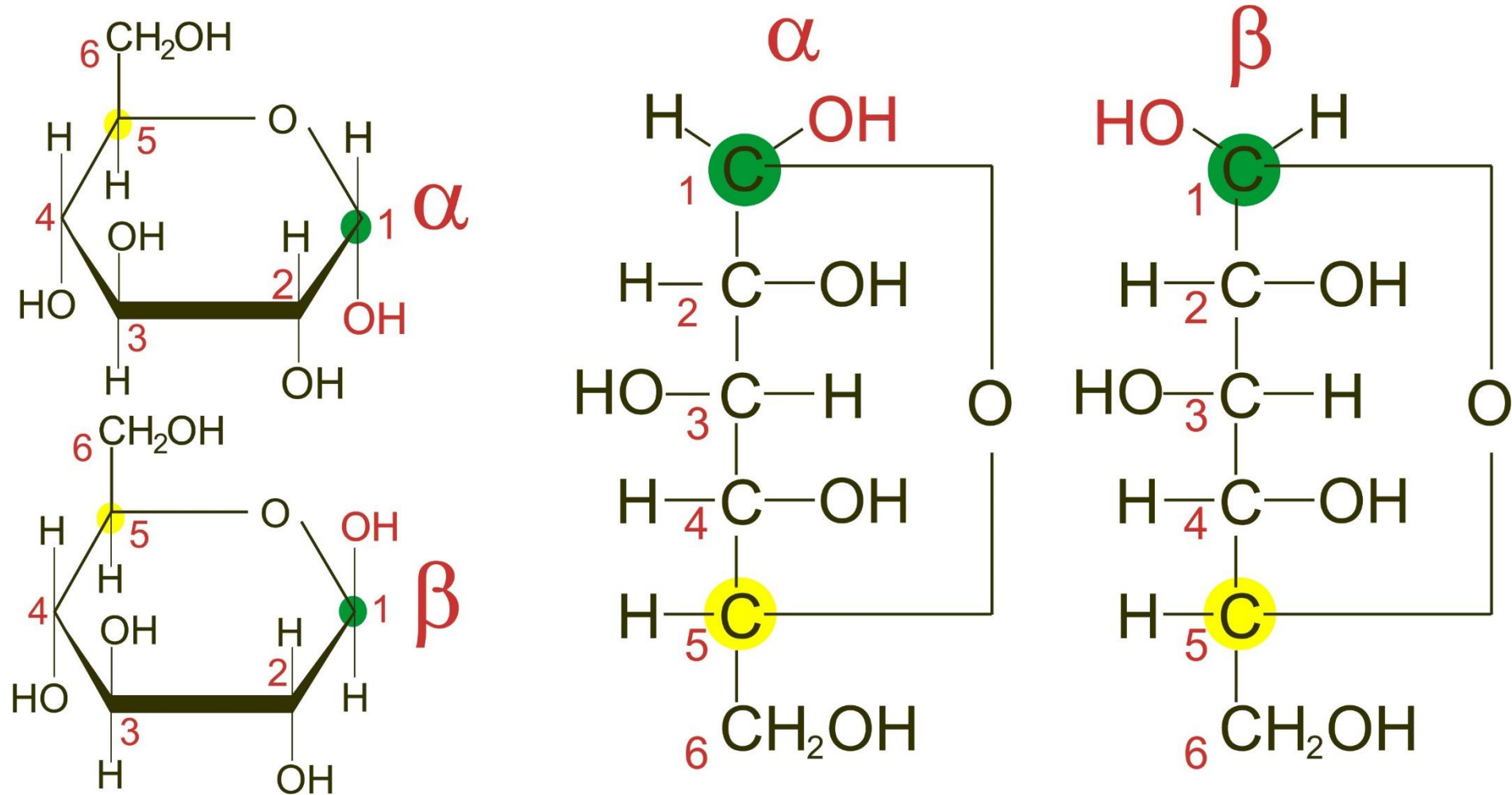
3- Epimers

The isomers formed due to variations in the configuration of $-H$ and $-OH$ around a single chiral carbon atom in a sugar molecule is called as epimers.



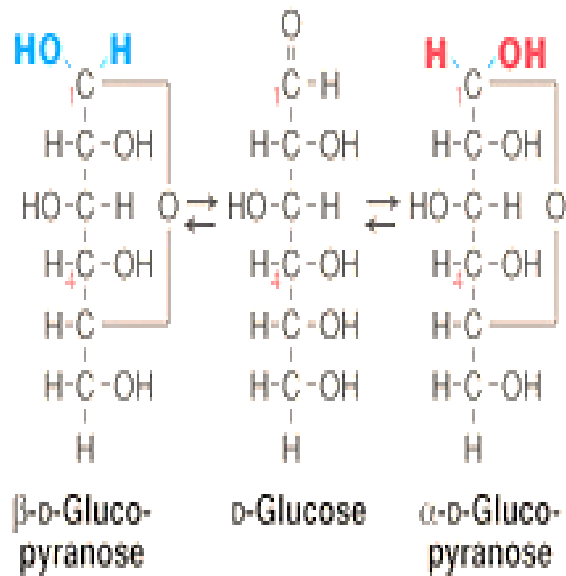
4- Anomers

Cyclization creates an anomeric carbon (former carbonyl carbon) generating the α and β configurations

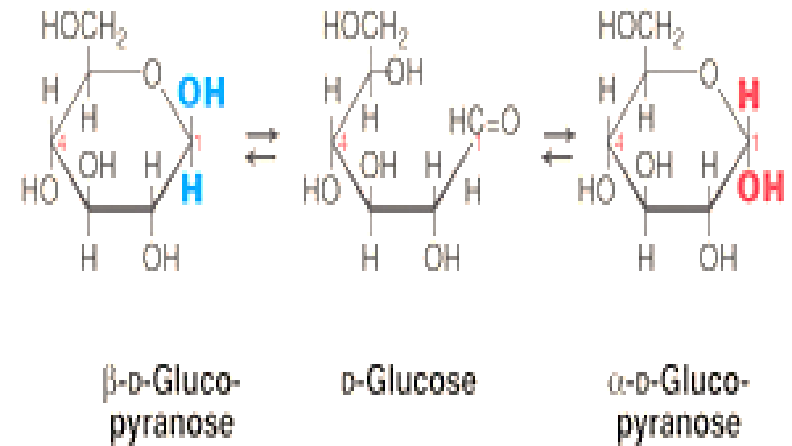


In solution, the cyclic α and β anomers of a sugar are in equilibrium with each other, and can be interconverted spontaneously.

A



B

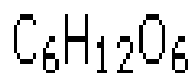


Pure α -D Glucose = specific rotation +112.2 °

Pure β -D Glucose = specific rotation +18.7 °

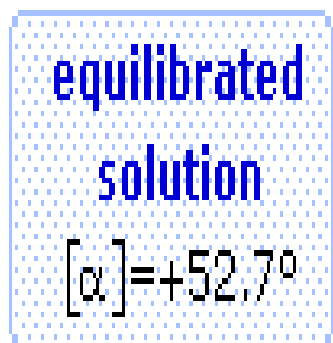
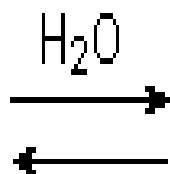
- ✓ **Mutarotation is a physicochemical property**
- ✓ **It is shown by certain optically active substances, who in aqueous solution has capacity to interconvert from one stereoisomeric form to other and attain a constant equilibrium mixture.**
- ✓ **This changes an initial specific rotation to a constant specific rotation with the passage of time.**

α -D-glucose

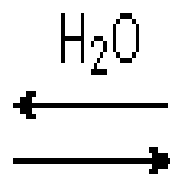


m.p. 146°

$[\alpha]=+112^\circ$



36% α + 64% β



β -D-glucose



m.p. 150°

$[\alpha]=+19^\circ$

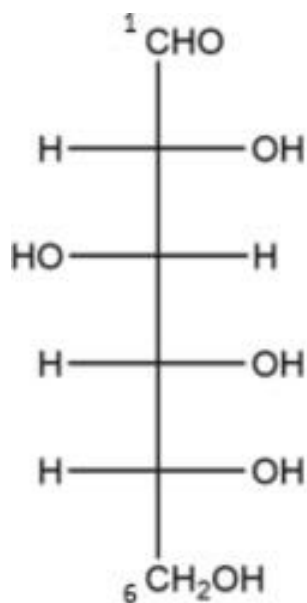
Physical properties of monosaccharide

- ✓ All monosaccharides are soluble in water.
- ✓ All monosaccharides have one or more chiral atom which can show the property of optical activity except dihydroxy acetone.
- ✓ All monosaccharides have isomers.
- ✓ Most monosaccharides occur in cyclic form which can exist in α and β forms.
- ✓ Most monosaccharides can undergo mutarotation.
- ✓ All monosaccharides are reducing sugar that can reduce Benedict reagent.

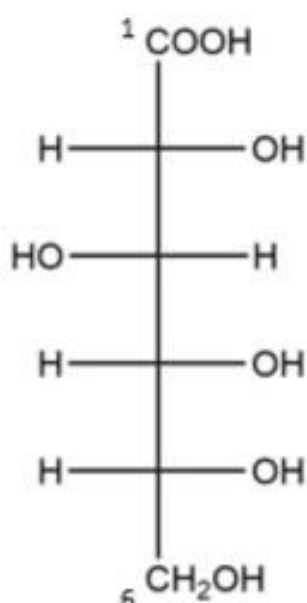
Chemical properties of monosaccharide

1- Oxidation.

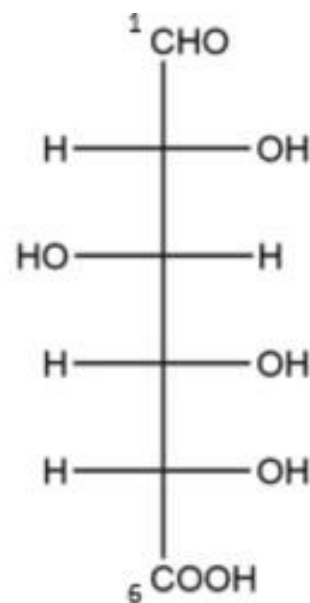
- A. Oxidation of aldehyde group gives aldonic acids e.g. glucose gives gluconic acid.
- B. Oxidation of the last alcoholic group gives uronic acids e. g. glucose gives glucuronic acid.
- C. Oxidation of both aldehyde and last alcoholic groups gives saccharic acids e. g. glucose gives glucosaccharic or glucaric acid.



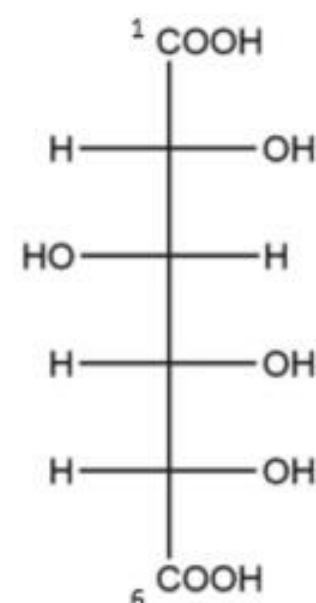
D-glucose



D-gluconic acid



D-glucuronic acid

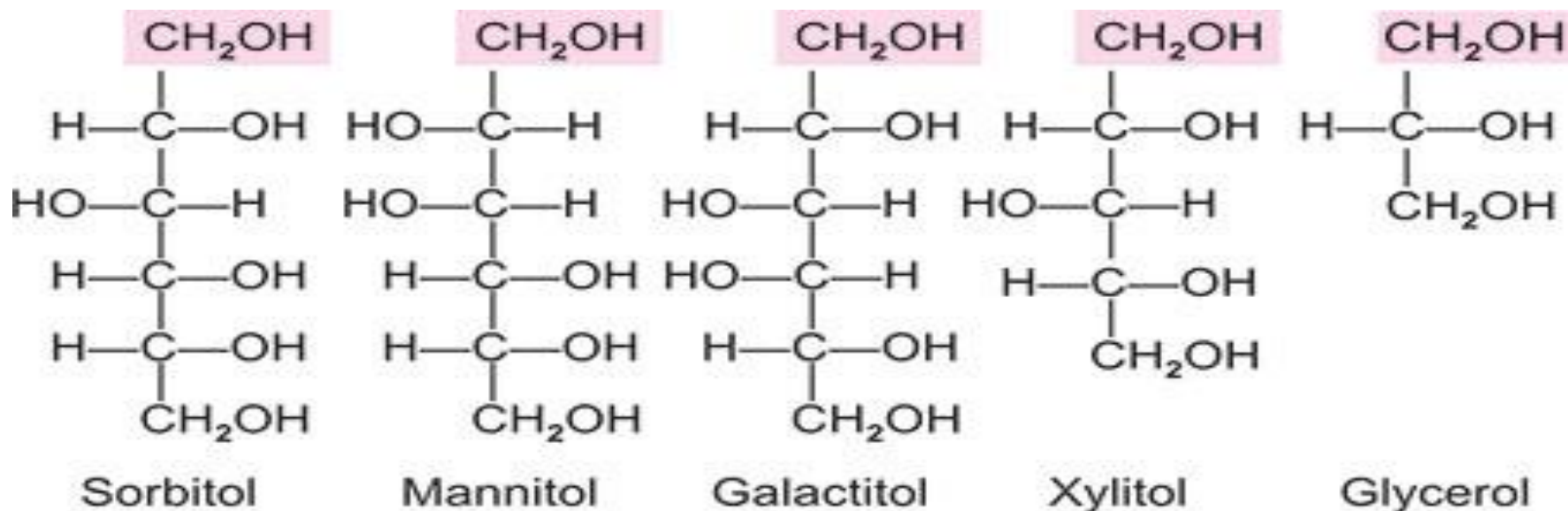


D-glucaric acid

Chemical properties of monosaccharide

2- Reduction.

- A. Reduction of aldehyde or ketone groups gives the corresponding alcohols i.e. glucose gives sorbitol, ribose gives ribitol, glyceraldehyde gives glycerol, galactose gives galactitol, mannose gives mannitol and fructose gives mixture of sorbitol and mannitol.



Chemical properties of monosaccharide

3- Action of alkali

- A. Weak alkali result in isomerization and interconversion of sugar e. g. glucose gives rise to a mixture of galactose, fructose and mannose.
- B. Strong alkalis result in polymerization of sugar molecules where a faint odor of caramel is produced when sugar is warmed in the presence of KOH or NaOH (Moore's test).

4- Action of acids

- A. Phosphoric acid gives phosphate esters e. g. glucose-6-phosphate and fructose-6-phosphate.
- B. Sulphuric acid: it is a dehydrating agent removes 3 molecules of water from the sugar giving a compound called furfural.

Chemical properties of monosaccharide

5- Fermentation.

It is the action of bacteria or enzymes of yeast on sugars gives rise to ethyl alcohol and CO_2 . All hexoses are fermented (D form only).

6- Osazone formation.

Osazones are yellow crystals compounds resulting from the reaction of sugars with phenylhydrazine. This reaction depends on the presence of free aldehyde or ketone groups thus all monosaccharides can form Osazone crystals.

Glucosazone, mannaosazone, fructosazone = insoluble bundle of grass

Galactosazone = insoluble needle shape

